

Navigation during Wildfire (Receiver Device)

The concept of the Fire Detection Egg is to provide real-time transmission of the fire status on the ground and transmit via Long Range (LoRa) if a fire is detected. At this point, the transmitter egg is only briefly described.

At this stage in the design process, the transmitter device is designed as a multi-walled ceramic egg:

Inside is a LoRa transmitter and a minimal memory with its own ID.

In the event of a fire, the electronics are powered by Peltier elements. These convert a temperature difference into a voltage. In order to maintain this effect for longer, an extinguishing foam capsule inside the egg is conceivable.

Once the power supply is established, the ID is sent out as a repeating message via LoRa.

How can the data then be used for our purpose?

A receiver device should be able to be used by hikers and firefighters. The design should be intuitive to use and robust with just a few physical buttons. The materials used should be fire and water resistant and durable over time.

Your own location and the fire locations are shown on the display. There is an arrow between each fire location and your own position, which changes color if the fire is visible in the displayed image section (red) or if it is not (orange). The arrows are annotated with the absolute compass direction and the relative distance to your own location. As soon as a fire is reported at a distance of less than 100 m, a warning appears on the display. "Top" on the display is in the direction of view.

The GPS coordinates of the transmitting eggs are available in a local database so that the receipt of an ID corresponds with the stored coordinates. This reduces the amount of data sent to a minimum and also the hardware in the transmitter egg. However, it also requires precise documentation of the positions and the immobility of the transmitter, which could be a problem in the forest due to curious people or animals or weather conditions.

Both devices are slim in their range of functions. This means they are well suited to their purpose and can be easily replaced or repaired.

The receiver device requires the following components:

Hardware Components

Component	Explanation
GPS-Sensor (e.g. https://www.iigs.uni-stuttgart.de/forschung/projekte/low-cost-gps/index.html Garmin eTrex)	Find own position
Digital Compass (Magnetometer) (e.g. https://www.bosch-sensortec.com/media/boschsensortec/downloads/datasheets/bst-bmm350-ds001.pdf)	Being able to quickly and precisely recognize your own orientation in order to intuitively assess the real direction of the flames. An analog compass element would be disturbed by the electronic components.
LoRa Receiver Module	LoRa has low energy consumption and a long sending distance. Since a high data rate is not required, LoRa is a good choice for this project.
Database with GPS-Locations of the Detection Eggs + Offline-Map of surroundings	Well-documented and statically stored ID to GPS location pairs can minimize the amount of ice transmitted. Contour lines, rivers and hiking trails should be shown on the map for better orientation.
Single-board computer (Raspberry Pi, Arduino, ...)	Execute the program code.

Software-Components:

Component	Explanation
LoRa Decoder	@In: LoRa Recv Data @Body: Präambel Detection, ggf. Hammingcode Correction or similar robustness, binary transmission codes to int @Out: ID (int)
GPS Decoder	@In: Received GPGLL data @Body: Save GPS data

Magnetometer Decoder	Rotate map and compass rotate depending on which direction you are facing.
GPS to LatLong	@In: GPS Location as GPGGA format @Out: Latitude and Longitude (exists)
ID to GPS Location	@In: ID @Out: GPGGA value (exists)
Calculate distance and direction	@In: own position, fire positions @Out: Annotated 2D image with arrows from own position to fire positions. Annotation includes distance and one of 16 compass directions (exists)
Update Interrupts	Map needs to be updated with the data of the magnetometer and the Fire Detection Eggs
Process buttoninputs	--
Righthand-/Lefthand-Mode	Software conversion of the button inputs, the magnetometer and the display orientation

Design:

- fireproof and waterproof, lightweight design with physical buttons for intuitive and gloved use
- concealed switch for switching between right-handed and left-handed use

The interactive map shows the direction in which the device is pointing, the contour lines(, rivers and hiking trails [Future Work]) in the surrounding area and your own location in relation to reported fire locations (Fig. 1).



Abbildung 1: Entwurf für das Receiver Device

How could we proceed?

The map display already shows the contour lines, but does not yet show rivers and hiking trails for better orientation.

The zoom and pan function is currently designed for a touchpad or mouse setup, but should be possible in the future with the physical buttons from the design in Figure 1.

A hardware prototype would be a possible next step in order to be able to write and test all decoders. This would also involve switching to button inputs instead of touchpad and mouse inputs.

In this first proof of concept, some performance weaknesses of the Python code are already visible. Possible improvements would be refactoring, an

alternative to displaying with Matplotlib or a change/fusion of the programming language(s).

In addition, regular updates of the direction based on the magnetometer data and further interrupts, for example when new LoRa data is received, are needed.

The possibility of deleting reported fires in the event of a malfunction or after the fire has been extinguished is also an important design element that should be considered in future iterations.